

Social Media Social Media and Networks Analytics

Assignment 2 : Choose Your Own Analysis

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1. Research Question

This project investigates how developers discuss trust and distrust in AI coding assistants, and how public discussion communities shape concerns around productivity, reliability, security, privacy, cost, code ownership, and future developer work.

The central research question is:

What drives trust and distrust in AI coding assistants among developers, and how do influential users and discussion communities shape concerns around productivity, reliability, security, cost, and code ownership?

2. Motivation

AI coding assistants have moved from experimental tools into everyday developer workflows. Tools such as GitHub Copilot, Cursor, Claude Code, OpenAI Codex, and other agentic coding systems are now discussed not only as productivity aids, but also as sources of technical risk, workflow disruption, privacy concern, and professional uncertainty. This makes the topic suitable for social media and network analysis because public discussion is both text-rich and relational: users comment on videos, participate in issue threads, cluster around products, and repeatedly discuss similar concerns across platforms.

The project is motivated by a practical question: if developers are expected to adopt these tools, what kinds of public evidence show where trust is being built or eroded? The analysis therefore focuses on discussion patterns rather than only product features. This allows the project to combine text analysis of comments and issue content with network analysis of users, repositories, videos, and discussion threads.

3. Data Sources

This project uses public online discussion data from three platforms: YouTube, GitHub, and Hacker News. These sources were selected because they capture different types of discussion about AI coding assistants and provide enough text data for sentiment and topic analysis, while also supporting basic network analysis.

YouTube provides broader public and developer-facing reactions through video comments and replies. GitHub provides product-specific discussion through issues and issue comments, including bug reports, feature requests, reliability concerns, and workflow problems. Hacker News provides technology-community discussion around AI coding tools, software development, automation, and the wider impact of AI on programming work.

Using multiple sources helps reduce reliance on a single platform or audience. YouTube comments capture public reaction around videos, GitHub issues capture user feedback in product communities, and Hacker News captures discussion from a technical news/community setting. Together, these datasets allow the project to compare how trust, distrust, productivity, reliability, cost, privacy, and job displacement themes appear across different online discussion environments.

4a. Data Collection Method - YouTube

YouTube data was collected using the YouTube Data API v3. The collection process followed the course-provided YouTube API workflow and retrieved videos related to AI coding assistants, including GitHub Copilot, Claude Code, OpenAI Codex, AI coding agents, vibe coding, developer productivity, privacy, security, and job displacement.

For each search query, the collection pipeline retrieved video metadata and top-level comments. The video metadata included title, channel, publication date, view count, like count, and comment count. The raw nested JSON files were retained for reproducibility, then flattened into comment-level tables for cleaning and analysis.

The unit of analysis for YouTube text analysis is a single comment. Each comment row retains associated video metadata, allowing results to be grouped by video, channel, search query, or publication period.

4b. Dataset Summary - Youtube

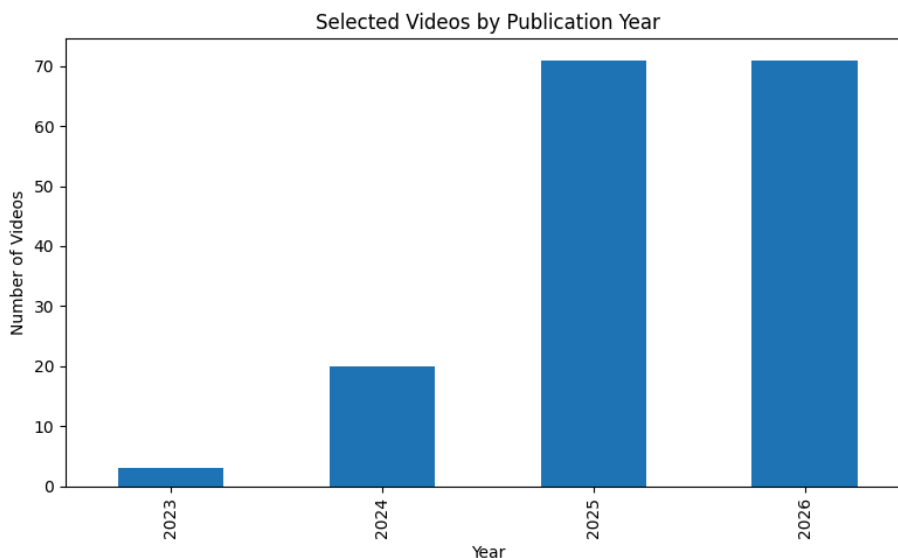
The final cleaned YouTube dataset contains 23,871 comments from 165 unique videos, written by 19,589 unique commenters. The dataset includes both top-level comments and replies, with 18,166 top-level comments and 5,705 replies.

The collected comment date range is 7 November 2023 to 22 May 2026. The selected videos cover AI coding assistant discussion across topics such as GitHub Copilot, Claude Code, OpenAI Codex, AI agents, privacy, security, productivity, vibe coding, and developer replacement.

4c. Data Cleaning - Youtube

The YouTube cleaning process removed duplicate comments, removed empty or missing comment text, converted date fields into datetime format, and created cleaned text fields for NLP analysis. The original comment text was retained alongside the cleaned text so preprocessing choices could be reviewed.

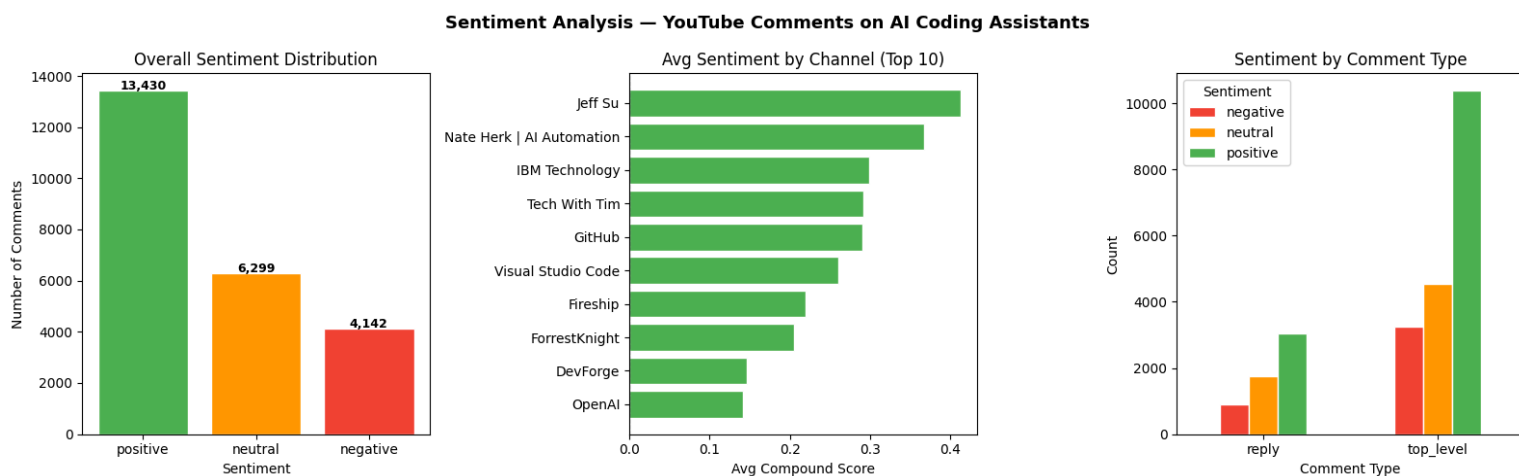
The cleaned text field standardises the comment text for sentiment analysis and topic modelling.



This allows the YouTube data to be compared with the GitHub and Hacker News datasets using a similar NLP workflow.

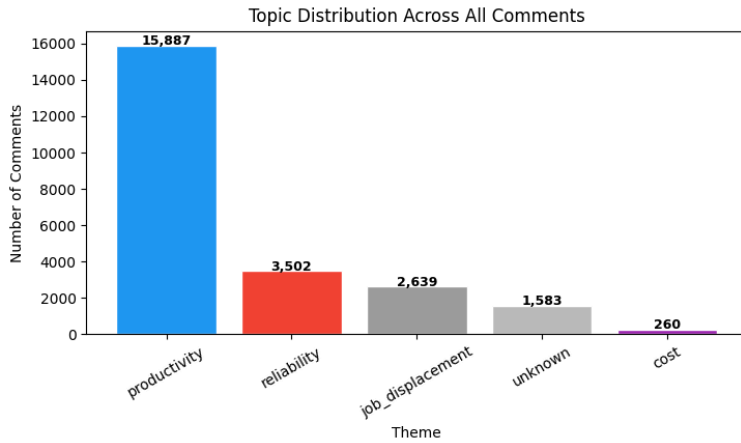
4d. Sentiment and Topic Analysis - Youtube

YouTube sentiment analysis was conducted using VADER. Each comment was assigned a compound sentiment score and classified as positive, neutral, or negative. This allows the YouTube discussion to be compared with GitHub using the same sentiment method.



Topic modelling was conducted using LDA on the cleaned YouTube comments. The model produced six learned topics, which were mapped into broader research themes. The strongest theme was productivity, followed by reliability and job displacement.

LDA Topic Modelling — AI Coding Assistant Comments

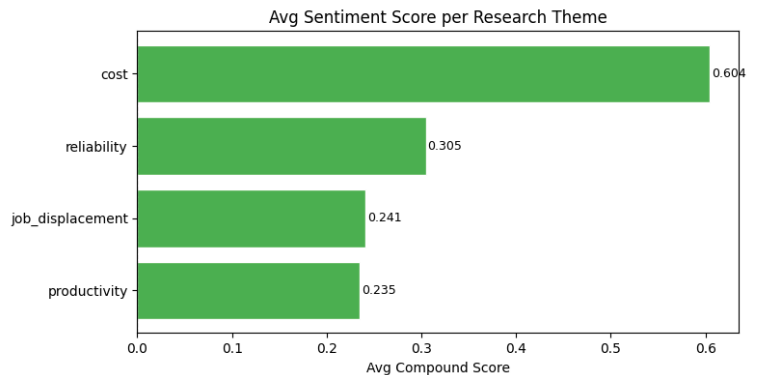
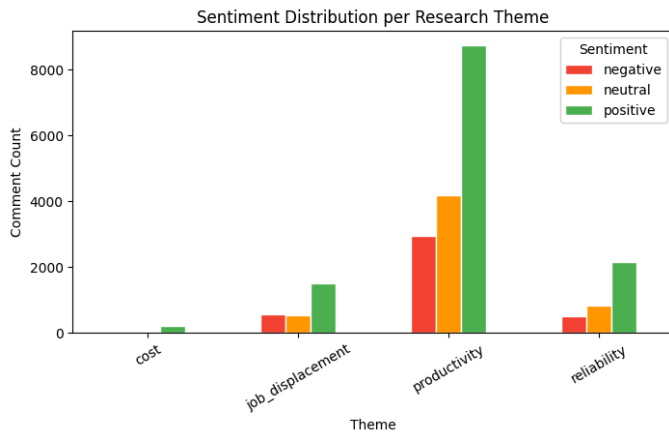


Top 8 Terms per LDA Topic

Topic 0 (job_displacement)	Topic 1 (productivity)	Topic 2 (reliability)	Topic 3 (reliability)	Topic 4 (cost)	Topic 5 (productivity)
software	time	agent	window	vibe	workflow
job	year	model	function	made	step
engineer	problem	never	error	look	content
developer	understand	project	check	yes	plan
company	learn	app	type	stuff	replacing
replace	day	prompt	crazy	bro	output
human	write	build	salary	video	spec
tool	junior	lim	close	free	say

The sentiment-by-topic output compares whether different themes have different emotional tones. In the YouTube dataset, all major mapped themes had positive average sentiment, but the topic breakdown still helps separate broad enthusiasm about AI coding tools from concerns around reliability, cost, and job displacement.

Sentiment × Topic Cross-Analysis



Channel-level topic comparison was also used to show whether different YouTube channels emphasised different parts of the AI coding assistant discussion.

5a. Data Collection Method - GitHub

GitHub data was collected using the public GitHub REST API from three public repositories associated with AI coding assistants: `microsoft/vscode-copilot-release`, `openai/codex`, and `anthropics/claude-code`. These repositories were selected because they contain product-specific user discussions, including bug reports, feature requests, reliability concerns, usability complaints, and feedback about AI-assisted coding workflows.

The collection focused on GitHub issues and issue comments. Pull requests were excluded because GitHub's issues endpoint can return both issues and pull requests, and this project is focused on public user discussion rather than code review activity.

For each repository, the collection pipeline retrieves recently updated public issues, excludes pull requests, collects public comments attached to each issue, saves raw JSON by repository, and then flattens the data into analysis-ready tables.

5b. Dataset Summary - GitHub

The current GitHub dataset contains 1,200 public issues and 6,150 public issue comments. After cleaning and NLP filtering, 5,828 discussion records remain for sentiment and topic modelling. The issue dataset includes both open and closed discussions, with 658 open issues and 542 closed issues. The collected issue date is currently from 20 June 2023 to 25 May 2026, and the comment date range is from 23 October 2023 to 25 May 2026.

5c. Data Cleaning - Github

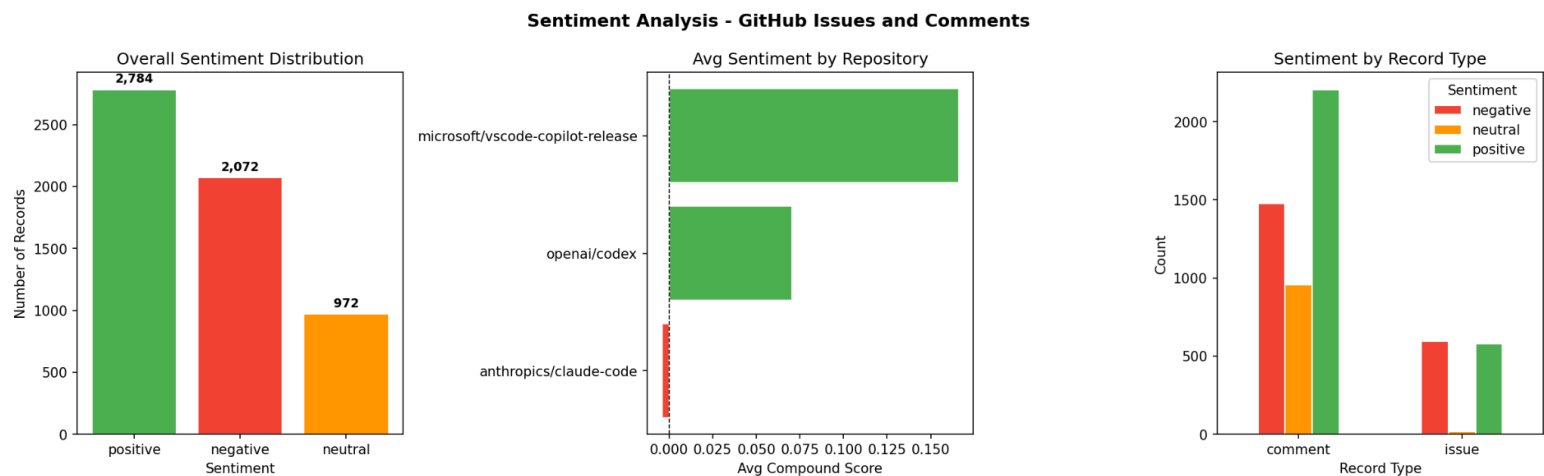
The GitHub cleaning pipeline keeps raw text and cleaned text separately. This is important because GitHub discussions often contain Markdown, code snippets, logs, URLs, stack traces, and issue template content. Removing all technical formatting too aggressively could remove meaningful context, so the raw Markdown text is retained for auditability while a cleaned text field is created for NLP.

The cleaning process removes empty text records, strips code blocks and inline code formatting, removes URLs and common Markdown characters, normalises whitespace, and creates lower-case cleaned text fields. After this first cleaning stage, 7,340 discussion-text records remained.

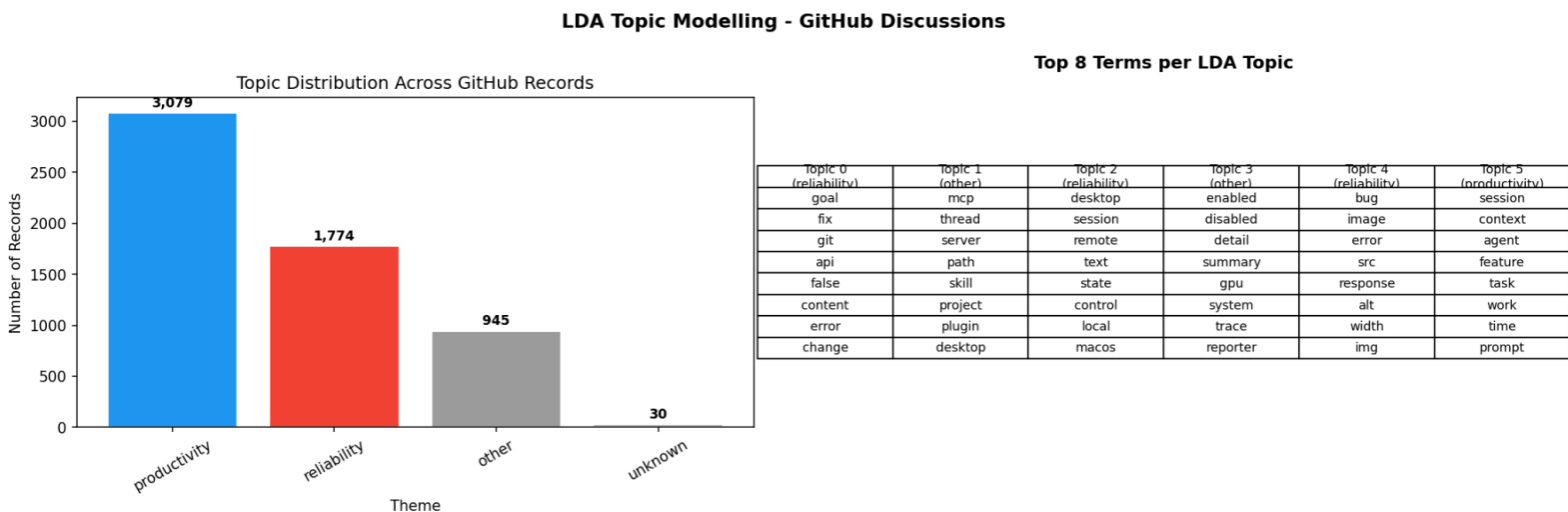
A second filtering stage is applied only for NLP. This removes rows that are unlikely to contribute meaningful sentiment or topic information, including bot-authored comments, automated duplicate/stale messages, generated/co-authored markers, very short cleaned text under 25 characters, pile-on comments such as "+1" or "same here", and very long comments over 3,000 characters that are likely to contain logs or templates.

5d. Sentiment and Topic Analysis - Github

GitHub sentiment analysis was conducted on the 5,828 filtered discussion records using VADER. Each issue or comment was assigned a compound sentiment score and classified as positive, neutral, or negative. The GitHub discussion dataset was mostly positive, but contained a substantial negative component, which is expected because GitHub issues often focus on bugs, failures, missing features, and technical problems.

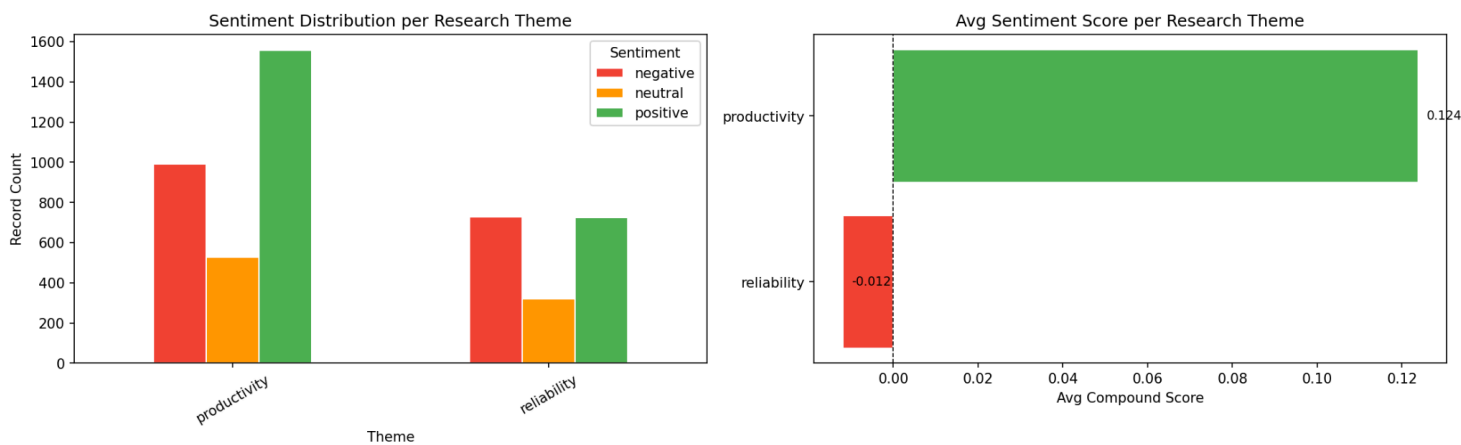


Topic modelling was conducted using LDA on the cleaned GitHub discussion text. The model produced six learned topics, which were mapped into broader research themes. The main themes identified in the GitHub data were productivity and reliability, with a smaller group of miscellaneous technical discussion.



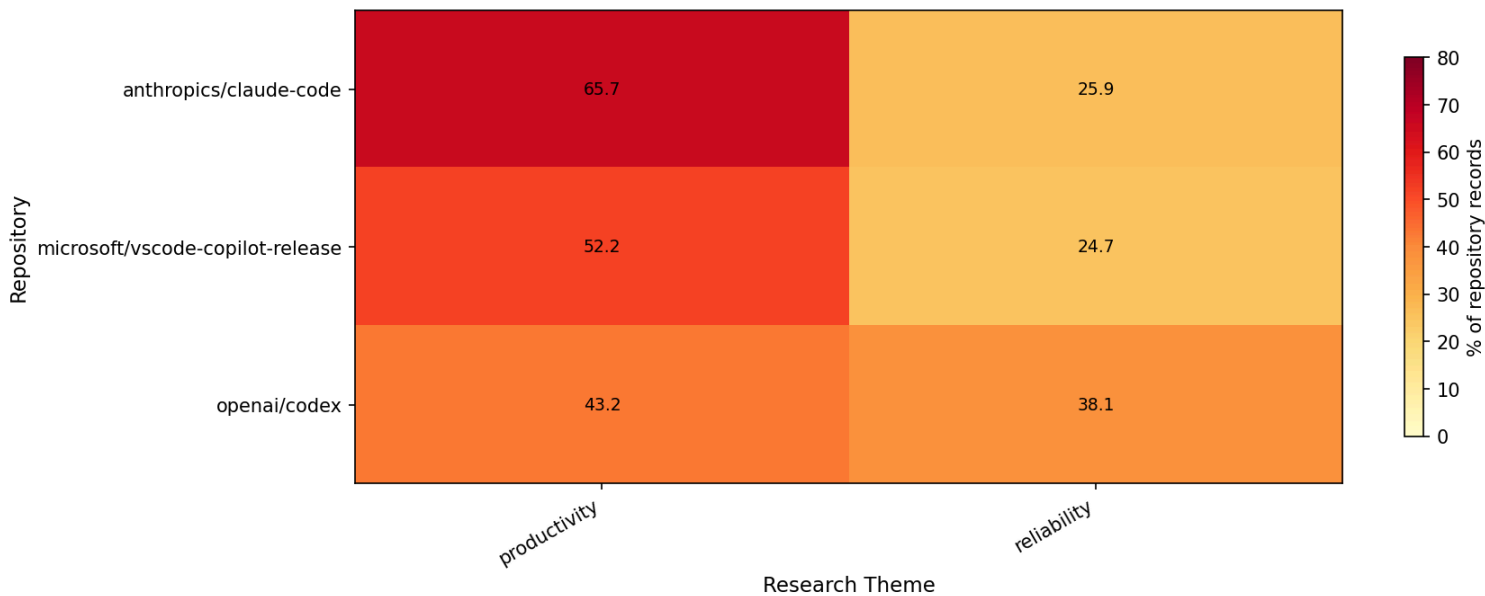
The sentiment-by-topic output shows how sentiment differs across the identified themes. Productivity-related discussion was more positive overall, while reliability-related discussion was more mixed and slightly more negative. This fits the GitHub context, where reliability themes often relate to bugs, errors, platform issues, failed commands, or broken workflows.

Sentiment x Topic Cross-Analysis - GitHub



Repository-level topic comparison shows that the three GitHub repositories have different discussion profiles. This helps compare whether user discussion around each AI coding assistant is mainly about productivity, reliability, or other technical concerns.

Topic Distribution by GitHub Repository (% of Records)



The word cloud is included as a supporting visual for common trust and distrust language in GitHub discussions. It should be interpreted with caution, because word clouds show frequent terms but not context or significance. This is especially important for GitHub, where technical terms like errors, files, sessions, and commands may appear often due to issue reporting. Therefore, the word cloud supports the sentiment and topic results but is not used as primary evidence.



6a. Data Collection Method - Hackernews

Hacker News data was collected using the public Hacker News APIs. The collection was done using Hacker News Firebase API to retrieve the full story and comment data for each thread meanwhile the Hacker News Algolia Search API was used to find the threads. The queries focused on tools and concerns related to AI assisted programming which includes AI coding agents like Claude, Github Copilot, OpenAI Codex, security and privacy issues and productivity of software developers.

For each query, the data collection retrieved stories and then collected metadata like story ID, title, author, date, comment count. The story ID was then used to retrieve public comments, author and other details. The raw results were saved in a JSON file and after that it was flattened into CSV files for cleaning and analysis.

6b. Data set Summary - Hackernews

The Hacker News dataset contained 222 story records and 3,111 comment records which gives us a total of 3,333 text records to analyse the text. The dataset is mainly comments which captures user interactions towards a certain post. The story records on the other hand gives us background/context about the thread.

The dataset was collected using 8 search queries "GitHub Copilot", "Claude Code", "OpenAI Codex", "AI coding assistant", "vibe coding", "AI replacing programmers", "AI code security

privacy" and "AI software developer productivity", this gave us a broad view of the discussions about tools available in the market and concerns related to AI assisted programming.

For network analysis, a basic user story edge table was created in which a Hacker News user is connected to a story if they comment on that story. The edge just shows how many times a user commented on the same story giving us a simple network.

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Hacker News dataset summary
```

```
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```

```
Number of stories: 222
```

```
Number of comments: 3111
```

```
Number of text records: 3333
```

```
Story date range: 2018-03-27 to 2026-05-23
```

```
Comment date range: 2018-03-27 to 2026-05-24
```

6c. Data Cleaning - Hackernews

The Data from Hacker news was cleaned to prepare it for sentiment analysis as well as topic modelling. The raw data contained links, encoded characters, white spaces and so these were removed or they were normalised before the analysis.

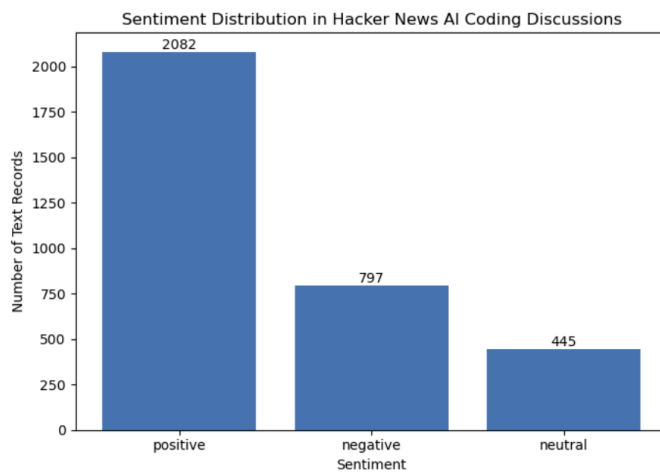
The cleaning pipeline kept the original story and comment information separately and a cleaned text field for NLP. Very short/ empty text records were removed because they do not provide any useful information for the analysis. The cleaned text was then converted to lowercase for tokenisation, stopwords were removed, lemmatisation, word clouds were made, VADER sentiment analysis and LDA topic modelling.

A basic network edge table was created during preprocessing as mentioned earlier. This table connects each Hacker News user to the story thread they commented on.

6d. Sentiment and Topic Analysis - Hackernews

Hacker News text analysis was carried out on the cleaned story and comment records using VADER sentiment analysis and LDA topic modelling. VADER was used to assign each text record a compound sentiment score and then it classifies it as positive, neutral or negative depending on the score. This gives a general view of whether Hacker News discussions around

AI coding assistants were supportive, critical or mixed.



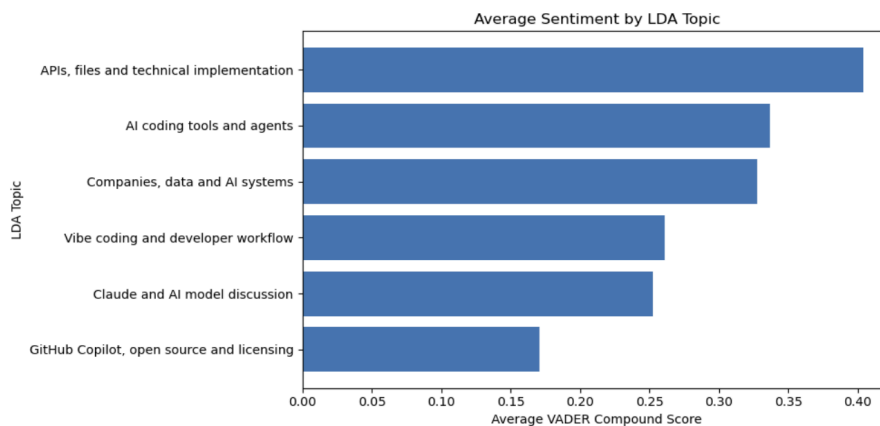
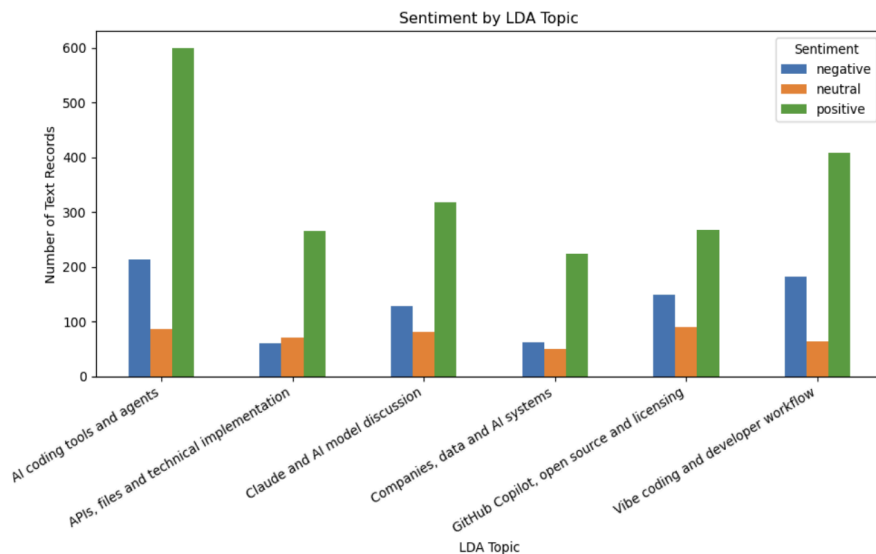
The overall sentiment was mostly positive, with 2,082 positive records, 797 negative records and 445 neutral records. The average compound sentiment score was 0.29, which suggests that discussion was generally positive but still contained a clear amount of criticism and concern. Positive comments often discussed usefulness, productivity and coding workflow improvements, while negative comments often focused on reliability, incorrect outputs, privacy, licensing or even concerns about over-reliance on AI tools. The positive and negative word clouds were used as supporting visuals to compare the language used in different sentiment groups.



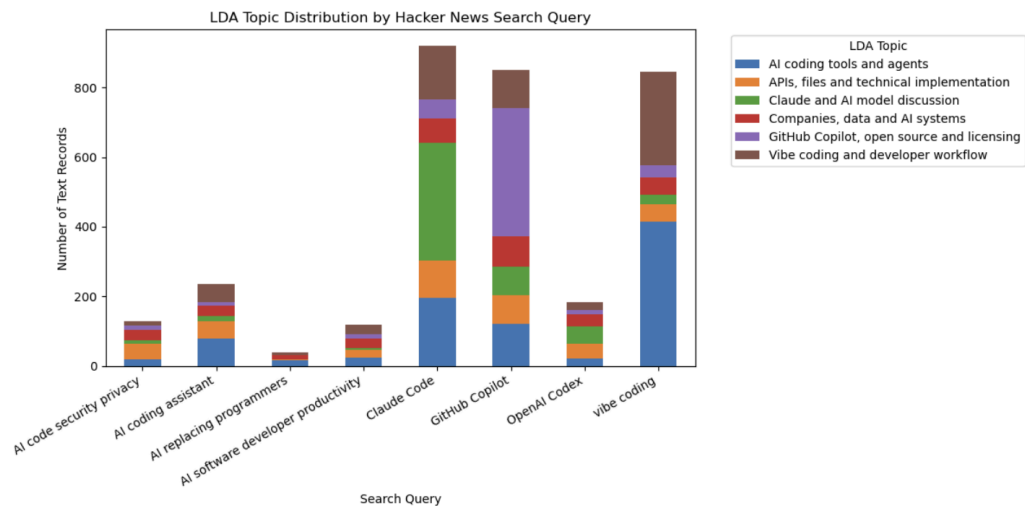
LDA topic modelling was then applied to the cleaned Hacker News text. The model used six topics. The largest topic was AI coding tools and agents with 899 records followed by Vibe coding and developer workflow with 654 records. Other major topics included Claude and AI model discussion with 527 records, GitHub Copilot, open source and licensing with 508 records, APIs, files and technical implementation with 396 records and Companies, data and AI systems with 338 records. This shows that Hacker News users discussed both specific AI coding tools and broader impacts on programming workflow.

	topicNumber	topWords	topicLabel
0	0	claude, code, model, plan, anthropic, token, u...	Claude and AI model discussion
1	1	data, system, product, href, company, model, g...	Companies, data and AI systems
2	2	code, llm, coding, tool, work, agent, write, v...	AI coding tools and agents
3	3	time, day, vibe, work, coding, lot, take, feel...	Vibe coding and developer workflow
4	4	code, copilot, github, source, open, license, ...	GitHub Copilot, open source and licensing
5	5	file, run, code, tool, model, api, data, user,...	APIs, files and technical implementation

Sentiment was also compared across the LDA topics. All topics had more positive than negative records, but the balance was different across topics. GitHub Copilot, open source and licensing had the lowest average sentiment score at 0.17 which suggests more mixed or cautious discussion. On the other hand, APIs, files and technical implementation had the highest average sentiment score at 0.40, suggesting that practical implementation discussion was more positive overall.



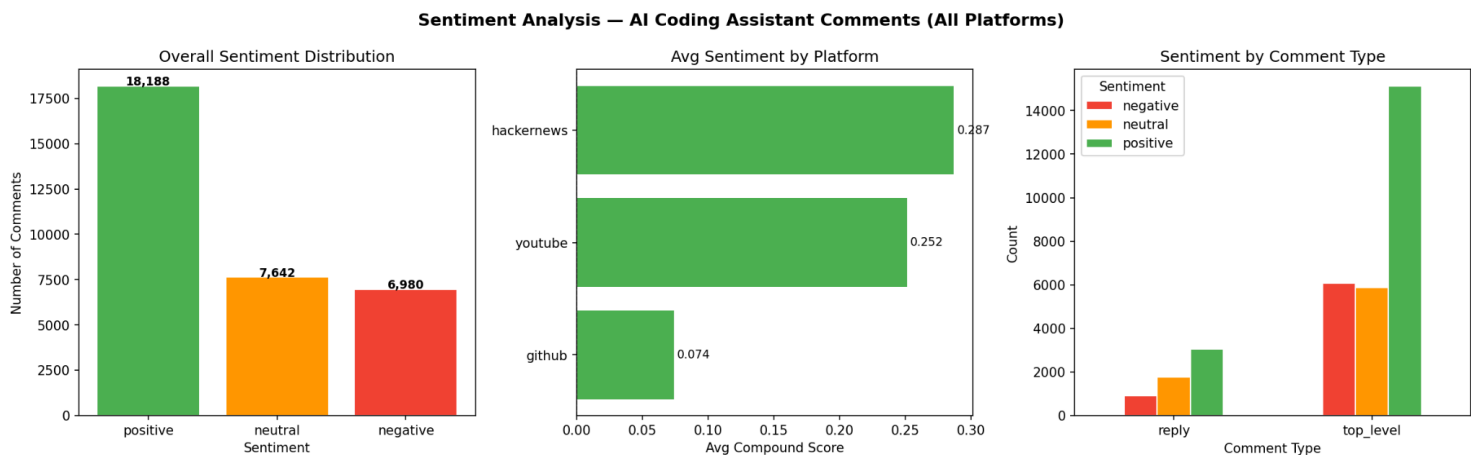
The LDA topic distribution by search query shows that different Hacker News search terms captured different areas of discussion. For example, the Claude Code query was strongly associated with Claude and AI model discussion, while the GitHub Copilot query was more connected to open source and licensing discussion. The vibe coding query captured more discussion about AI coding tools and developer workflow. This shows that the dataset covers multiple angles of developer trust and distrust instead of focusing on only one tool or issue.



7. NLP Comparative Analysis

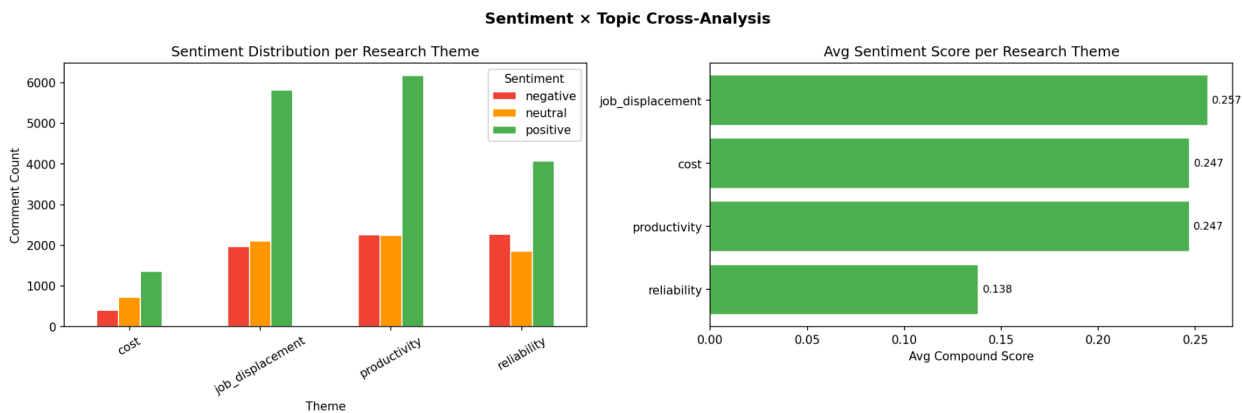
The comparative NLP analysis combines the cleaned YouTube, GitHub, and Hacker News datasets into a single comment-level file containing 32,810 records. This allows sentiment and topic results to be compared across platforms using the same VADER sentiment process and LDA topic-modelling pipeline.

Overall sentiment was mostly positive, with 18,188 positive comments, 7,642 neutral comments, and 6,980 negative comments. The average compound sentiment score was positive across all three platforms, but GitHub had the lowest average sentiment score at 0.0743, compared with YouTube at 0.2516 and Hacker News at 0.2871. This difference is expected because GitHub issues are often used to report bugs, errors, missing features, and reliability problems, while YouTube and Hacker News include more general discussion, tutorials, opinions, and reactions.

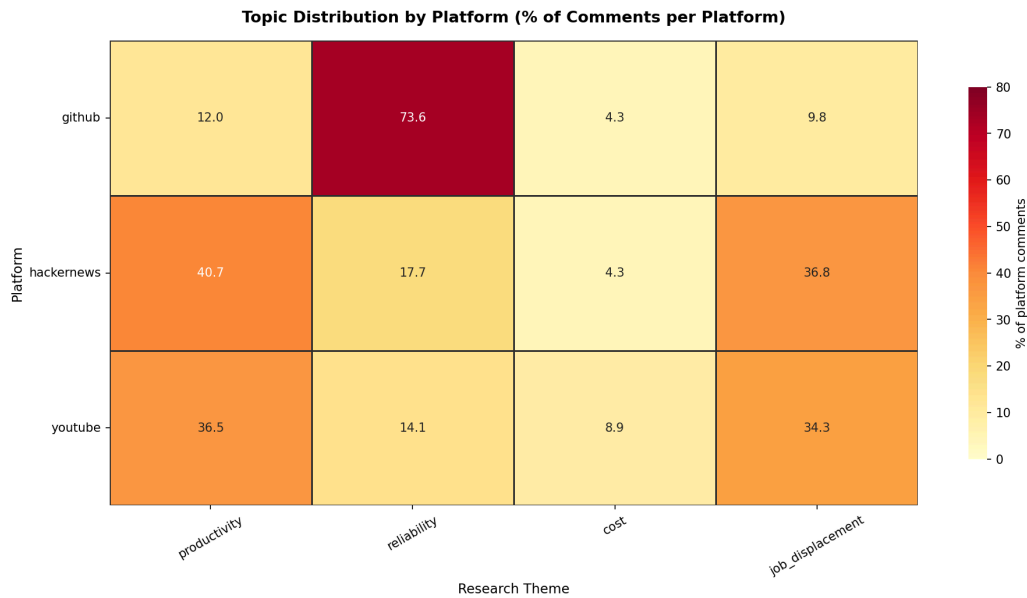


The topic model identified five main output categories in the combined dataset: productivity (10,680 comments), job displacement (9,896), reliability (8,204), cost (2,502), and unknown (1,528). Productivity, job displacement, and reliability therefore dominate the combined discussion. This supports the project focus, as developer trust appears to be shaped mainly by whether AI coding assistants improve workflow efficiency, whether they threaten or change developer work, and whether they are reliable enough to use in practice.

The sentiment-by-topic results show that all major topics had positive average sentiment overall, but reliability had the lowest average score at 0.1383. This suggests that reliability discussion is more mixed and more likely to contain distrust, frustration, or criticism than productivity-focused discussion. Productivity and job displacement both had stronger positive averages, but these should still be interpreted carefully because positive sentiment scores do not necessarily mean full trust; comments can be enthusiastic, sarcastic, speculative, or mixed in meaning.



The topic-by-platform heatmap shows strong platform differences. GitHub discussion is heavily reliability-focused, with 73.6% of GitHub records assigned to reliability. YouTube is more distributed across productivity (36.5%), job displacement (34.3%), and reliability (14.1%). Hacker News is concentrated around productivity (40.7%) and job displacement (36.8%). This indicates that each platform captures a different part of the broader trust and distrust conversation.

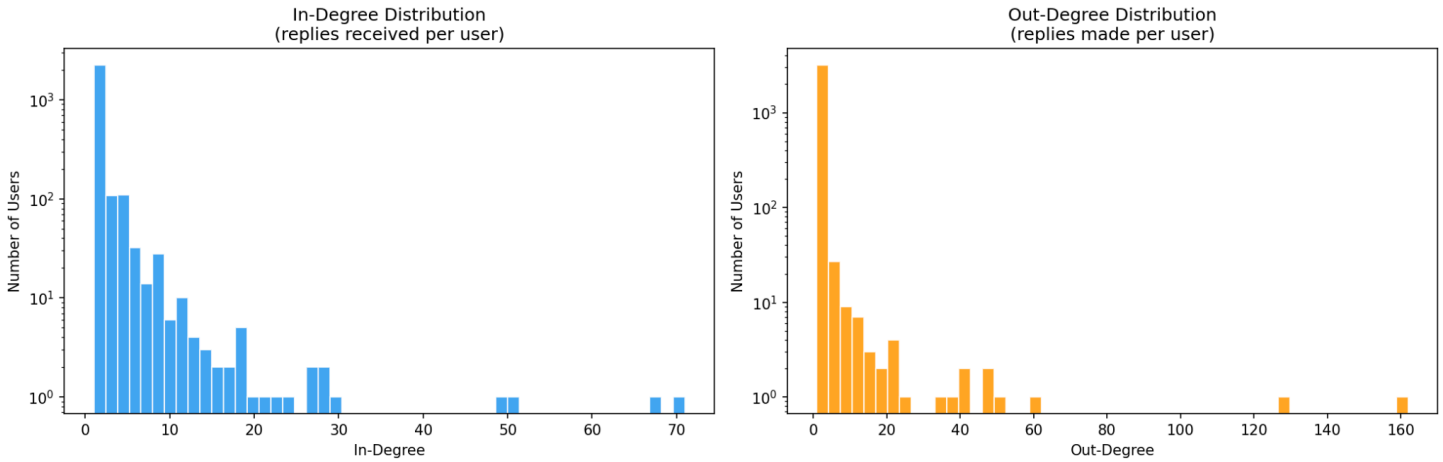


Network Analysis

The network analysis uses the combined edge dataset containing 14,128 raw edges. Two network structures were used. First, a directed reply graph was built from direct reply edges, mainly from YouTube and Hacker News. This graph is used for PageRank, in-degree, and centrality analysis because edge direction shows who replied to whom. Second, a broader community graph was built by combining reply edges with projected user-thread participation edges from GitHub and Hacker News. This allows GitHub issue participation to contribute to community detection even though GitHub issue comments are not direct replies.

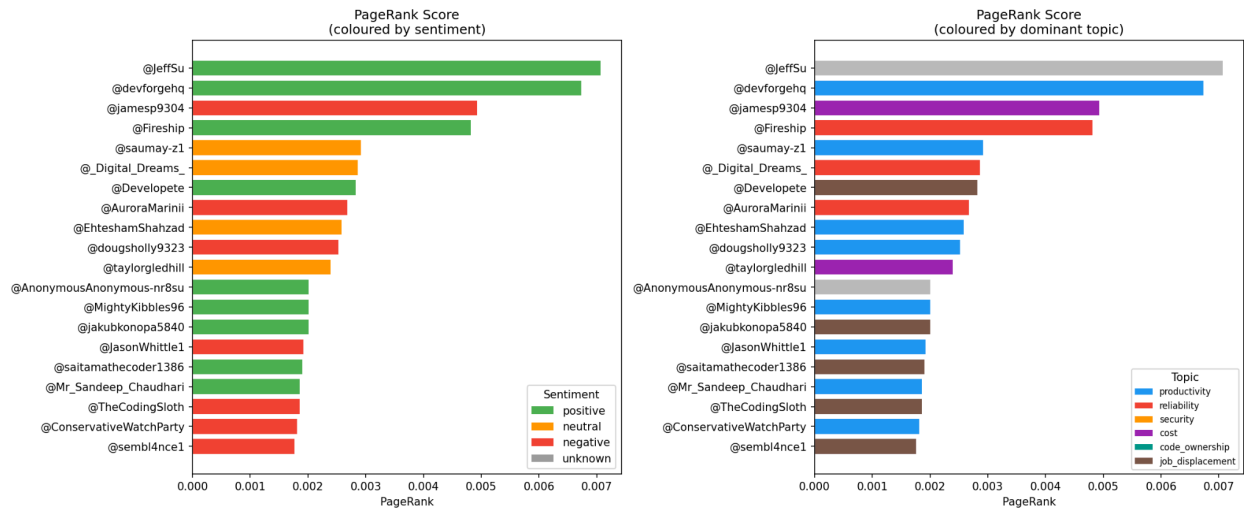
After basic edge cleaning, the directed reply graph contained 5,650 nodes and 4,766 edges, with a density of 0.000149. This low density is typical of social media discussion networks, where most users interact within isolated comment threads rather than across the whole dataset. The largest connected component contained 1,229 nodes, or 21.8% of the graph.

Network Degree Distribution



The centrality results show that a small number of users receive a disproportionate amount of attention. The top PageRank users include @JeffSu, @devforgehq, @jamesp9304, @Fireship, and @saumay-z1. Notably, @jamesp9304 appears as a highly central user with a negative

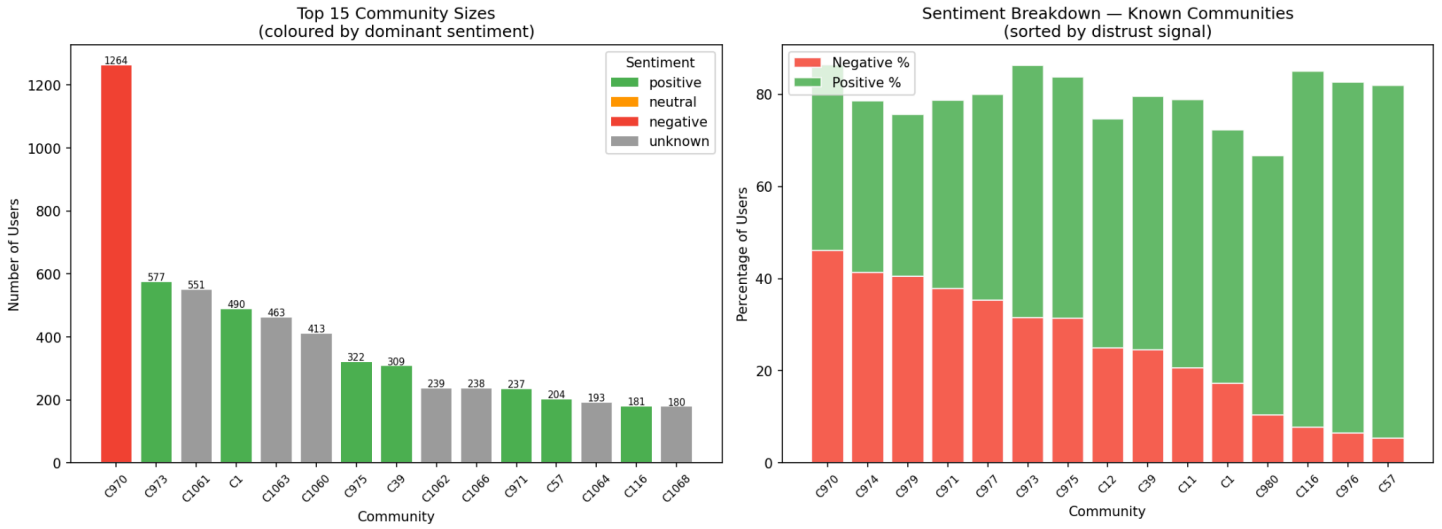
Top 20 Most Influential Users — Centrality Analysis



sentiment profile, showing that distrust-oriented comments can still occupy influential positions in the discussion network.

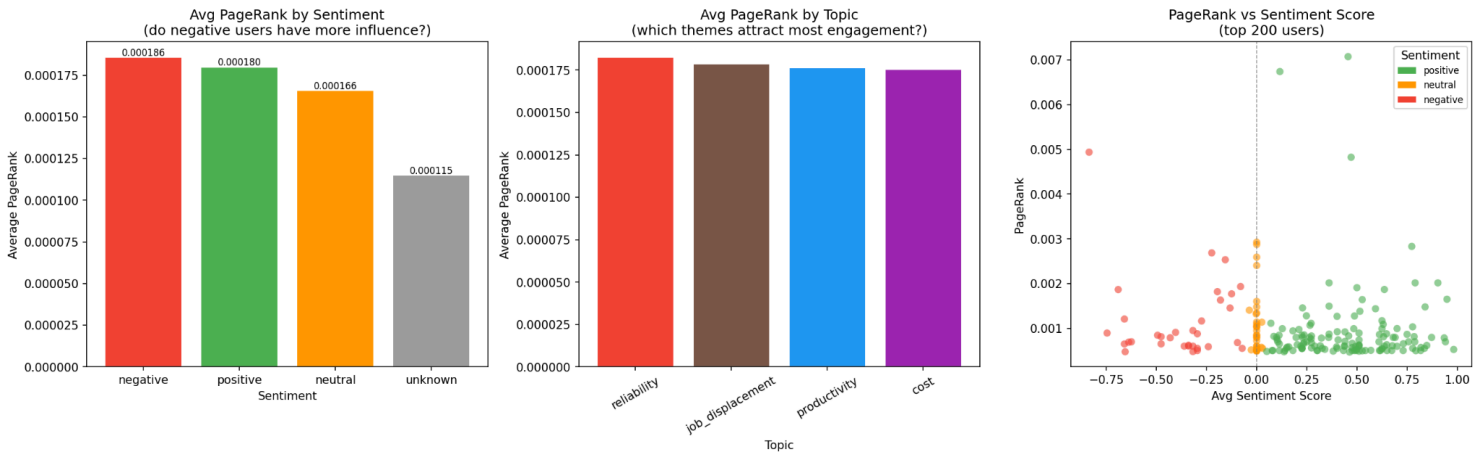
For community detection, the broader community graph contained 12,054 nodes and 126,825 edges after projecting shared thread participation into user-user connections. Louvain community detection identified 1,090 communities with a modularity score of 0.8657. This high modularity reflects the thread-based structure of the data: users tend to cluster around specific videos, issues, or stories rather than forming one unified cross-platform conversation.

Community Structure — AI Coding Assistant Discussion Network



The largest community had 1,264 users and was reliability-dominant with negative sentiment slightly outweighing positive sentiment. Other large communities were more positive and focused on reliability or job displacement. This suggests that distrust is not evenly distributed across the whole network, but appears more strongly in particular discussion clusters.

Network Position × Sentiment × Topic — Key Findings



Limitations

The YouTube dataset skews toward enthusiast audiences, inflating positive sentiment relative to developer forums. Search queries may have over-represented tutorial content at the expense of critical commentary. The LDA topic model was constrained to six topics, and security, cost, and code ownership themes were underrepresented across all platforms. More targeted data collection from licensing or privacy-focused communities would likely surface these themes more clearly. The Hacker News dataset produced only 22 directed reply edges due to incomplete

thread trees in the Algolia search results, limiting HN's contribution to the directed centrality analysis. Cross-platform user identity matching was not attempted, meaning a developer active on both YouTube and GitHub is treated as two separate nodes. This limits the accuracy of cross-platform influence findings. The Louvain modularity score of 0.9554 reflects the thread-local structure of social media reply graphs rather than meaningful ideological communities. Most detected communities correspond to individual video or story threads rather than user groups with shared perspectives.

Ethics and Data Handling

All data collected in this project is publicly available through official platform APIs and complies with the terms of service of each platform. No private, sensitive, or credential data was collected or stored at any point during the project.

YouTube data was collected using the official YouTube Data API v3 under the standard public quota allocation of 10,000 units per day. The total quota consumed across all collection runs was approximately 1,264 units, well within the daily limit. GitHub data was collected using the public GitHub REST API from three public repositories with publicly visible issues and comments. Hacker News data was collected using the public Hacker News Firebase API and the Algolia Search API, both of which are freely accessible without authentication for read operations.

API keys and access tokens were stored in a local `.env` file excluded from the project repository through `.gitignore` configuration. No API credentials were committed to version control at any point during the project. The `.env.example` file in the repository documents the required variable names without exposing any values.

All users whose comments appear in the dataset are identified by their public platform usernames or display names as returned by the respective APIs. No attempt was made to de-anonymise users, link accounts across platforms, or collect personally identifiable information beyond the public display names associated with API responses. Comment text used in the analysis was collected as-is from public responses and was not modified or misrepresented.

The analysis focuses on aggregate patterns - sentiment distributions, topic proportions, and network-level measures - rather than targeted analysis of individual users. The top influential users identified by PageRank are named in the network analysis section because they are identified by publicly visible usernames associated with high-volume public activity on public platforms, and their presence in the findings is a direct result of their own public participation. Raw data files were not committed to the project repository due to file size considerations and platform terms of service restrictions on data redistribution. A representative sample of the data structure is included in `data/samples/` as required by the assessment specification. All code required to reproduce the full dataset collection is included in the submitted notebooks.

Conclusion

This project investigated what drives trust and distrust in AI coding assistants among developers, and how influential users and discussion communities shape concerns around productivity, reliability, security, cost, and code ownership.

The analysis combined 32,810 comments from YouTube, GitHub, and Hacker News with a network of 14,128 edges across 25,630 unique users. Sentiment analysis using VADER and topic modelling using LDA identified productivity and reliability as the two dominant themes of developer discussion about AI coding tools. Positive sentiment is associated primarily with productivity-themed discussion, while negative sentiment is concentrated in reliability-themed discussion and job displacement concerns.

The network analysis of 5,650 nodes and 4,766 directed reply edges identified a small set of highly influential users who attract disproportionate community engagement. These influential users are not uniformly positive - the presence of high-PageRank users with negative sentiment profiles, particularly on YouTube, confirms that distrust-oriented commentary is capable of occupying structurally influential positions in developer discussion networks.

Platform context shapes how concern is expressed. YouTube surfaces job displacement concerns from a broad audience. GitHub surfaces reliability concerns from active product users. Hacker News surfaces technical scepticism from an experienced developer audience. This platform specialisation means the relative visibility of different concern types depends on which communities a developer participates in, and no single platform provides a complete picture of the trust and distrust landscape.

Security, cost, and code ownership concerns were underrepresented in the collected data relative to the research question's framing. This is itself a finding: at the time of data collection, developer discourse about AI coding assistants is dominated by productivity and reliability, with legal and privacy concerns receiving comparatively little attention in the mainstream communities studied here.

The combined findings suggest that trust in AI coding assistants is conditional and context-dependent. Developers express trust when discussing productivity gains and express distrust when confronting reliability failures, professional displacement, and the limitations of current tools. Influential users amplify both signals, and the communities they participate in determine which concerns receive the most visibility in public discourse.

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